



Green Roof Report

Suryajal · AI rooftop solar + rainwater assessment

Report #C4138EA8

26 Sep 2026

NASA POWER climatology



YOUR ROOF

**7th B Main Road, Geetha Colony,
Bengaluru**

12.92852° N, 77.58213° E

216 m² (2,325 sq ft roof)

Usable for solar: 151 m² (70% of roof)

Roof type: RCC / concrete flat roof (runoff 0.85)

Outline: AI - MobileSAM (Segment Anything), confidence 1.00

Panel layout: 5 × 540 Wp, auto-placed with 0.5 m edge gap



GREEN SCORE

Solar 60/60 · Water 40/40



Rooftop Solar

2.70 kW · 5 panels · sized by your electricity use · 4.2 units/day per kW here

SYSTEM SIZE

2.70 kW

5 × 540 Wp panels

UNITS PER YEAR

4,185

≈ 349 units / month

BILL SAVINGS / YEAR

₹26,697

≈ ₹2,225 per month

COST AFTER SUBSIDY

₹1.03 lakh

₹1.75 lakh — ₹72,600 PM Surya Ghar

PAYBACK

3.9 years

then ~21 years of free power

25-YEAR SAVINGS

₹6.29 lakh

net gain ₹5.26 lakh

CO₂ AVOIDED

3.0 t / year

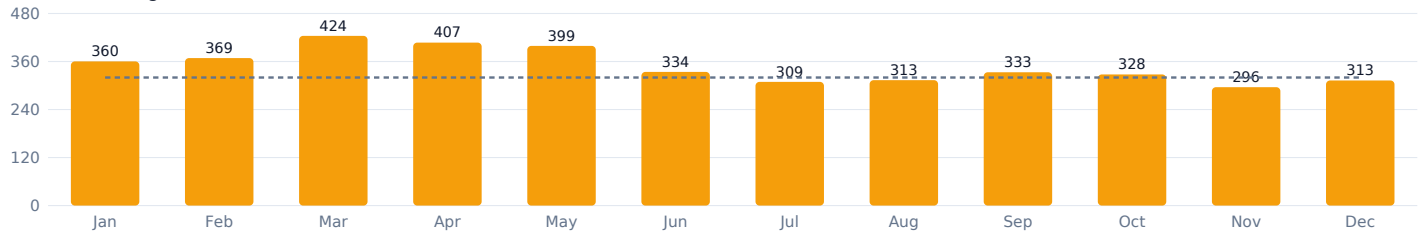
70 t over 25 years

LIKE PLANTING

149 trees

at ~20 kg CO₂ per tree per year

Month-wise solar generation (units)



Rainwater Harvesting

843 mm rain/year · harvest = area × rain × 0.85

RAINWATER PER YEAR

1.55 lakh L

from 843 mm of rain

WATER FOR

229 days

family of 5 at 135 L/person/day

STORAGE TANK

10,000 L

captures a 50 mm rain day

RECHARGE WELL

1 m Ø × 6.0 m

holds 4,712 L (BWSSB 20 L/m²)

BWSSB MINIMUM

4,320 L

storage or recharge by law (Bengaluru)

TANKERS AVOIDED

26 / year

of 6,000 L each

WATER MONEY SAVED

₹20,627

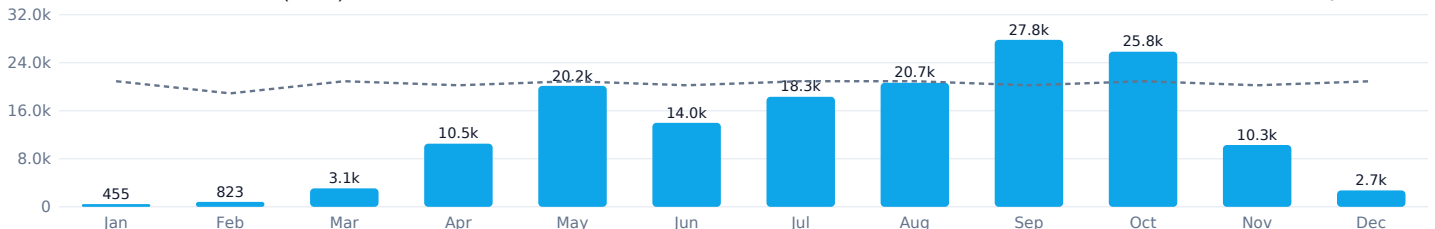
at ₹800 per tanker

WETTEST MONTH

Sep

27,819 L in that month

Month-wise rainwater harvest (litres)



How we calculated

Solar: NASA POWER sunlight (5.49 kWh/m²/day) × temperature-corrected performance ratio 0.77; 10 m² per kW (PM Surya Ghar portal); 540 Wp panels.

Money: ₹65,000/kW installed; subsidy ₹30k/kW up to 2 kW + ₹18k for the 3rd kW (max ₹78k); tariff ₹6.82/unit; surplus exported at ₹2.14/unit (KERC 2026).

CO₂: CEA grid factor 0.71 kg/unit; panels lose 0.5%/year; tariff held flat (conservative).

Rain: 1 mm on 1 m² = 1 litre; runoff 0.85; tank sized for a 50 mm rain day; recharge well sized to BWSSB's 20 L per m² of roof.

This is a screening estimate for awareness, not an engineering design. Before buying, get a site survey from an MNRE-empanelled vendor and apply on pmsuryaghar.gov.in.

Climate: NASA POWER climatology 2001-2020 (satellite + MERRA-2). Imagery © Esri, Maxar, Earthstar Geographics & GIS User Community. Made with Suryajal (MobileSAM + FastAPI).



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